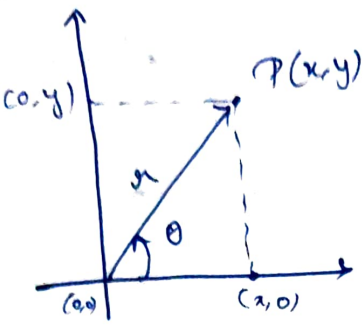
Differential Calculus.

1) Polar Curves :-



$$x = r\cos\theta \rightarrow (1)$$
$$y = r\sin\theta \rightarrow (2)$$

$$\sin\theta = \frac{y}{r} \qquad \cos\theta = \frac{x}{r}$$

Squaring and adding eq^n (1) & (2)

$$x^2 + y^2 = r^2(\sin^2\theta + \cos^2\theta)$$
$$\Rightarrow x^2 + y^2 = r^2$$
$$\Rightarrow \boxed{r = \sqrt{x^2 + y^2}}$$

Dividing (2) ÷ (1),

$$\frac{y}{x} = \frac{r\sin\theta}{r\cos\theta}$$
$$\Rightarrow \tan\theta = y/x$$
$$\Rightarrow \boxed{\theta = \tan^{-1}\left(\frac{y}{x}\right)}$$

Q] $P(x,y) \equiv (3,3)$

In Polar

$\rightarrow r = \sqrt{(3)^2 + (3)^2}$

$= \sqrt{18}$

$= 3\sqrt{2}$

$(x,y) \equiv (3,3)$

$x \equiv 3, y \equiv 3$

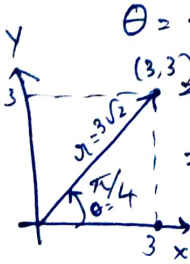
$r = \sqrt{x^2+y^2}$

$\theta = \tan^{-1}\left(\frac{y}{x}\right)$

$= \tan^{-1}\left(\frac{3}{3}\right)$

$= \tan^{-1} 1$

$= \frac{\pi}{4}$



$\therefore P(3\sqrt{2}, \frac{\pi}{4})$

Q) $P(r,\theta) \equiv P(\sqrt{2}, \pi/4)$

$\rightarrow \theta \equiv \pi/4$, $r = \sqrt{2}$

$\tan\theta = \frac{y}{x}$

$1 = y/x$

$y = x$

$P(1,1)$

$\sqrt{2} = \sqrt{2x^2}$

$\sqrt{2} = x^2\sqrt{2}$

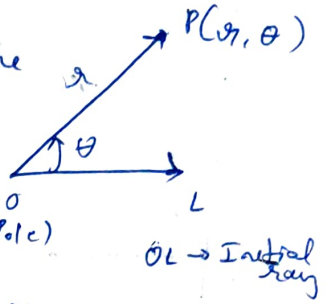
$x = 1 = y$

⊙ **POLAR CO-ORDINATES:-**

The pair (r, θ) represented by $P(r, \theta)$ is called the polar co-ordinates of the point P.

⊙ **POLE:-**

The initial reference point O on the plane is called the pole



⊙ The line OL drawn through O is called the initial ray.

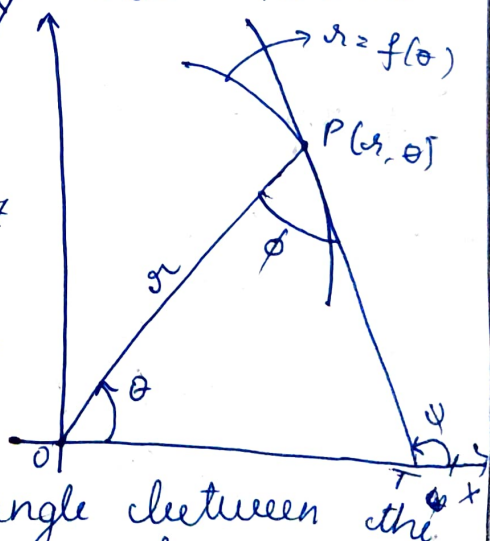
⊙ The line joining O & P at distance r is called the radius vector.

⊙ The angle LOP = θ is called vectorial angle and is always in anticlockwise direction.

POLAR CURVES:- A Polar Curve is a graph described by an equation in the polar co-ordinates instead of Cartesian Co-ordinates and it is given by $r = f(\theta)$.

***ANGLE BETWEEN RADIUS VECTOR AND TANGENT:-**

Let $P(r, \theta)$ be any point on the polar curve $r = f(\theta)$. Let PT be the tangent drawn to the curve at P such that it meets the x-axis at T. Let ψ be the angle made by the tangent with the +ve axis measured in anti-clockwise direction. Let ϕ be the angle between the radius vector OP and the tangent vector PT.



Consider $\triangle OPT$,

By exterior Angle Property,

$$\psi = \theta + \phi$$

Take tan on both sides

$$\tan\psi = \tan(\theta+\phi)$$

$$\boxed{\tan\psi = \frac{\tan\theta + \tan\phi}{1 - \tan\theta\tan\phi}} \rightarrow (1)$$

$x = r\cos\theta$ & $y = r\sin\theta$ $\Rightarrow$ $x = f(\theta)\cos\theta$, $y = f(\theta)\sin\theta$

$$\frac{dy}{d\theta} = f'(\theta)\sin\theta + \cos\theta f(\theta)$$

$$\frac{dx}{d(\theta)} = f'(\theta)\cos\theta - \sin\theta f(\theta)$$

$$\tan\psi = \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

(Slope of tangent)

$$\tan\psi = \frac{f'(\theta)\sin\theta + \cos\theta f(\theta)}{f'(\theta)\cos\theta - \sin\theta f(\theta)} \times \frac{\frac{1}{f'(\theta)\cos\theta}}{\frac{1}{f'(\theta)\cos\theta}}$$

(Multiply & dividing by $f'(\theta)\cos\theta$)

$$= \frac{\frac{f'(\theta)\sin\theta}{f'(\theta)\cos\theta} + \frac{f(\theta)}{f'(\theta)}}{\frac{f'(\theta)\cos\theta}{f'(\theta)\cos\theta} - \frac{\sin\theta f(\theta)}{f'(\theta)\cos\theta}}$$

$$\tan\psi = \frac{\tan\theta - \frac{f(\theta)}{f'(\theta)}}{1 - \tan\theta\frac{f(\theta)}{f'(\theta)}}$$

$$\boxed{\tan\psi = \frac{\tan\theta + \frac{f(\theta)}{f'(\theta)}}{1 - \tan\theta\frac{f(\theta)}{f'(\theta)}}} \rightarrow (2)$$

Comparing ① & ②

$$\frac{\tan\theta + \tan\phi}{1 - \tan\theta\tan\phi} = \frac{\tan\theta + \frac{f(\theta)}{f'(\theta)}}{1 - \tan\theta\frac{f(\theta)}{f'(\theta)}}$$

$$\tan\phi = \frac{f(\theta)}{f'(\theta)}$$

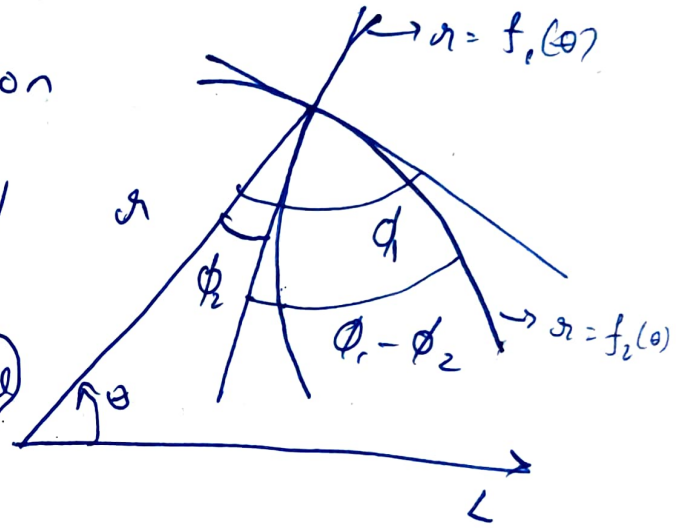
$$= \frac{r}{\frac{dr}{d\theta}}$$

$$\tan\phi = r\cdot\frac{d\theta}{dr} \rightarrow \phi = \tan^{-1}\, r\frac{d\theta}{dr}$$

$$\cot\phi = \frac{1}{r}\frac{dr}{d\theta}$$

★ Angle Of Intersection Of Two Curves :-

The angle of intersection of two curves is $\phi = |\phi_1 - \phi_2|$



$|\phi_1 - \phi_2| = \frac{\pi}{2} \Rightarrow$ curves are orthogonal or curves intersect orthogonally

↳ Angle of intersection

Consider, $\phi_1 - \phi_2 = \pi/2$

$\phi_1 = \pi/2 + \phi_2$

Take tan on both sides

$\tan\phi_1 = \tan(\pi/2 + \phi_2)$

$\tan\phi_1 = -\cot\phi_2$

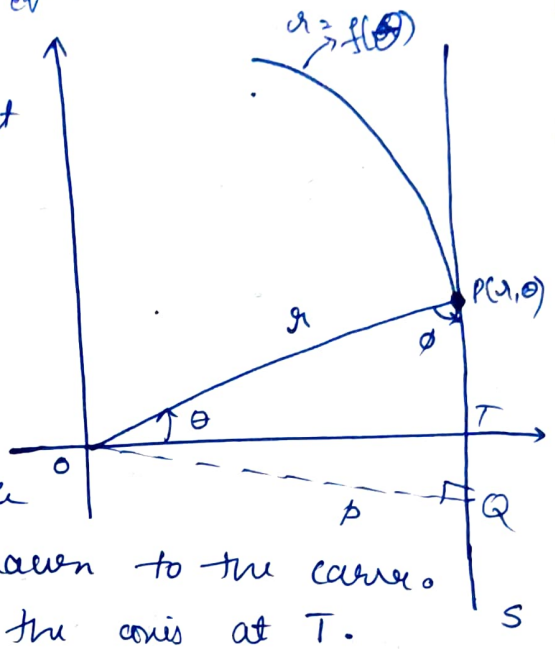
$\tan\phi_1 \tan\phi_2 = -1$

When curves are orthogonal i.e angle between $= \pi/2$

Hence curves are orthogonal.

⇒ PEDAL'S EQUATION :- (P.r eq^n)

* $\angle PTO \neq 90°$



Construction :-

Let P(r,θ) be the point at which the tangent is drawn to the curve $r = f(\theta)$. Let O be the pole. Let the distance OP be r. Let the radius vector OP make an angle ϕ with the tangent drawn to the curve. Let the tangent intersect the axis at T. Angle made by radius vector OP with the axis in the anticlockwise direction be θ. Extend the tangent PT to S, draw a perpendicular to tangent from point O, which intersect the tangent at Q at a distance 'p' from O.

Derivation :-

Consider, $\triangle OPQ$,

$$\sin\phi = \frac{OQ}{OP}$$

$$= \frac{p}{r}$$

$$\Rightarrow \boxed{p = r\sin\phi}$$ → form of Pedal's Eq^n

Taking reciprocal on both sides

$$\Rightarrow \frac{1}{p} = \frac{1}{r\sin\phi}$$

$$\Rightarrow \frac{1}{p} = \frac{\operatorname{cosec}\phi}{r} \quad \left(\because \frac{1}{\sin\phi} = \operatorname{cosec}\phi\right)$$

Squaring on both sides

$$\Rightarrow \left(\frac{1}{p}\right)^2 = \frac{\operatorname{cosec}^2\phi}{r^2}$$

$$\Rightarrow \frac{1}{p^2} = \frac{1+\cot^2\phi}{r^2} \quad (\because \operatorname{cosec}^2\phi = 1+\cot^2\phi)$$

$$\Rightarrow \frac{1}{p^2} = \frac{\left[1 + \left(\frac{1}{r}\frac{dr}{d\theta}\right)^2\right]}{r^2} \quad \left(\because \cot\phi = \frac{1}{r}\frac{dr}{d\theta}\right)$$

$$\Rightarrow \boxed{\frac{1}{p^2} = \frac{1}{r^2}\left[1 + \frac{1}{r^2}\left(\frac{dr}{d\theta}\right)^2\right]} \rightarrow \text{PEDAL'S EQUATION}$$

✡ RADIUS OF CURVATURE :-

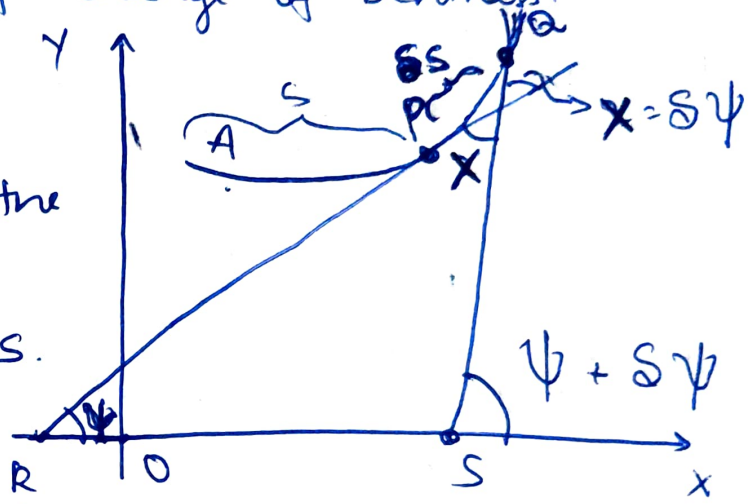
→ Curvature of a Curve :- Curvature of a curve is a measure of rate of change of bentness.

Construction :-

→ Let 'P' be any point on the given curve $r = f(\theta)$ and Q be the neighbouring point.

Let arc AP = s, and PQ = δs.

Let the tangents make the angles ψ and $\psi + \delta\psi$ with x-axis such that angle between the two tangents P & Q is $\delta\psi$.



while moving from 'P' to Q through a distance δs, the tangent has turned through the angle '$\delta\psi$'. This is called total bending or total curvature of the arc PQ

∴ The average curvature of arc

$$PQ = \frac{\delta\psi}{\delta s}$$

∴ Curvature at Point P is equal to,

$$\lim_{\Delta s \to 0} \frac{\psi + \delta\psi - \psi}{s + \delta s - s}$$

$$= \lim_{\delta s \to 0} \frac{\delta\psi}{\delta s} = \frac{d\psi}{ds}$$

$$\boxed{\frac{d\psi}{ds} = \lim_{\delta s \to 0} \frac{\delta\psi}{\delta s}}$$

It is denoted by K.

Radius of Curvature :-

The reciprocal of the curvature of the curve at any point P is called the radius of curvature at the point P. It is denoted by ρ, and is given by $\rho = \frac{ds}{d\psi}$

Different forms of radius of curvature :-

i] Cartesian Curve of the form $y = f(x)$

$$\boxed{\rho = \frac{\left[1 + (y_1)^2\right]^{3/2}}{y''}}$$

ii] Parametric Equations of the form,

$x = f(t)$

$y = g(t)$

$$\boxed{\rho = \frac{\left[(x')^2 + (y')^2\right]^{3/2}}{x'y'' - y'x''}}$$

iii] Polar curve of the form $r = f(\theta)$,

$$\boxed{\rho = \frac{\left[r^2 + (r')^2\right]^{3/2}}{r^2 + 2(r')^2 - rr''}}$$

⊙ Taylor's Series :-

$$f(x) = f(a) + \frac{(x-a)}{1!} f'(a) + \frac{(x-a)^2}{2!} f''(a) + \cdots\cdots \infty$$

⊙ Maclaurin Series. (Case of Taylor Series when $a = 0$)

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \cdots \infty$$